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How to Engineer Continuous Non-Invertible Symmetries

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Table of Contents

Foundations of Generalized Symmetries

The Conceptual Inversion of Logic

- **Traditional Noether Paradigm:**
  i) Continuous transformation $\phi \to \phi'$
  ii) Conserved current $\partial_\mu j^\mu = 0$, i.e. $d \star j = 0$
  iii) Charge Operator $Q = \int_\Sigma *j$
  iv) Symmetry operator $U_g(\Sigma) = \exp\{igQ(\Sigma)\}$

- **Modern Topological Paradigm:** [Gaiotto et al., 2015]
  We invert the logical arrow: **Every topological operator defines a symmetry.**

- **Advantages:**
  i) Continuous and discrete symmetries are treated uniformly
  ii) No reference to field transformations, then Nöether and Topological symmetries are on the same footing

First Generalization: Higher-Form Symmetries

Traditional symmetries act on local operators (0-dimensional points). Higher-form symmetries extend this action to non-local extended objects.

The Higher-Form Generalization:

- A p-form symmetry has charged objects of dimension p.
- The corresponding topological symmetry operator $U_g(\mathcal{M}_{d-q-1})$ has dimension $d - p - 1$.

Second Generalization: Non-Invertible Symmetries

Traditional symmetries form a group: every element $g \in G$ possesses a unique inverse g^{-1} such that $U_g \times U_{g^{-1}} = \mathbb{I}$.

The Non-Invertible Generalization:

- Topological operators do not necessarily admit an inverse.
- The fusion of two symmetry operators is of the form:

$$U_A(\mathcal{M}) \times U_B(\mathcal{M}) = \sum_C \mathcal{Z}_{AB}^C(\mathcal{M})\, U_C(\mathcal{M})$$

Here, $\mathcal{Z}_{AB}^C(\mathcal{M})$ represents the partition function of a TQFT living on $\mathcal{M}$.

The Algebraic Infrastructure: Higher-Fusion Categories

When we have generalized symmetries, in general, group theory fails to encode the symmetry structure of the theory and then is replaced by Higher-Fusion Categories.

- **Layered Hierarchy:** Symmetries are layered by their spatial dimensions. Each layer has a fusion rule.

- **Inter-Layer Connections:** Symmetries are not isolated; defects of a given dimension act as the morphisms (the maps) between the defects of the upper layer. They can occur as interface.

Physical Applications of Generalized Symmetries

Generalized symmetries, as Ordinary Symmetries, place exact, non-perturbative constraints on quantum field theory dynamics:

- **Selection Rules:** Forbid specific transitions in correlation functions.
- **RG Anomaly Matching:** Symmetries can have anomalies, which remain invariant under the Renormalization Group (RG) flow, constraining IR phases.
- **Generalized Landau Paradigm:** Classifies phases via the Spontaneous Symmetry Breaking (SSB).
- **EFT and Hydrodynamics:** Constrains effective actions at zero- and finite-temperature, and at equilibrium and slightly not.
- **Web of Theories:** Maps connections between distinct physical theories by tracking the explicit gauging of symmetries.

Example 1: The 2D Ising Model & Selection Rules

The 2D critical Ising model provides an elegant, discrete example of a non-invertible symmetry through the Kramers-Wannier duality operator [Shao, 2024].

- Symmetry operators: the invertible $\mathbb{Z}_2$ spin-flip η,
  and a non-invertible duality defect line $\mathcal{D}$.
- The Fusion Rules are :

$$\eta \times \eta = \mathbb{I}, \quad \eta \times \mathcal{D} = \mathcal{D}, \quad \mathcal{D} \times \mathcal{D} = \mathbb{I} + \eta$$

- The new defect acts as: $\mathcal{D}\epsilon = -\epsilon \qquad \mathcal{D}\sigma \sim \mu$

Physical Consequence: The defect $\mathcal{D}$ constrains certain correlation functions of the thermal ϵ, the order σ and disorder μ operators.

This selection rules cannot be explained using ordinary group representations.

Example 2: Axion Decay in QCD (ABJ Anomaly)

In 4D Quantum Electrodynamics with a massless Dirac fermion, the axial $U(1)_A$ symmetry is broken by the Adler-Bell-Jackiw (ABJ) anomaly.

- The current is dynamically non conserved: $d \star j_A = \# \, f \wedge f$.
- **Solution [Cordova and Ohmori, 2022, Choi et al., 2022]:** By *dressing* the naive axial operator with a 3D fractional Chern-Simons theory, one constructs a continuous non-invertible symmetry operator.

Physical Consequence: The presence of this (non-invertible) symmetry explains:

i) why helicity of (massless) electron-positron is conserved.

ii) why we have the therm $\int \pi^0 \, f \wedge f$ in the low energy EFT (of QCD), which allows the observed decay of the pion/axion into two photons ($\pi^0 \to \gamma\gamma$). [Choi et al., 2022]

Example 3: Magnetohydrodynamics as a 1-Form EFT (in $4D$)

Hydrodynamic: EFT governing the long-distance/time fluctuations of conserved charge densities. It is entirely characterized by its symmetries. [Grozdanov et al., 2017]

- **1-Form Global Symmetry:** Defined by the conservation of magnetic flux:

$$\star j^{(2)} = f^{(2)} \implies d \star j^{(2)} = 0$$

- **MHD Definition:** It is the study of electrically conducting fluids, such as plasmas, liquid metals, and salt water. The IR limit of $E\&M$ coupled to light charged matter at finite T.

Physical Consequence: 1-form Ward identities define universally resistivity.

First Take Home Message

The examples teach us a lesson:

We should care about Generalized Symmetries as much as we care about Ordinary Symmetries.

The Symmetry TFT Framework

Limitations of Background Field Formalism

Standard Framework: Couple background gauge fields B to global current j via

$$\mathcal{Z}[B] = \int \mathcal{D}\Phi\, e^{-S[\Phi]}\, e^{i\int B\wedge * j}$$

- **Conservation equation:** $\mathcal{Z}[B + d\Lambda] = \mathcal{Z}[B] \iff d \star j = 0$
- **'t Hooft & Mixed Anomalies:** Break this invariance under background gauge transformations.

$$\mathcal{Z}[B + d\Lambda] = e^{\mathcal{A}[\Lambda, B]} \times \mathcal{Z}[B]$$

- **Restore the invariance:** Couple the D-dimensional QFT to a $(D+1)$-dimensional anomaly theory.

Precious Information: Anomalies are not inconsistencies, but RG flow-invariant data.

The limitation: B is a group-valued field, it is not enough any more.

The SymTFT Sandwich Construction

To decouple the symmetry data from dynamical degrees of freedom, we extend background fields to a $(d+1)$-dimensional topological bulk (as for the anomaly theory).

- **Bulk ($\mathcal{Y}_{D+1}$):** Houses the Symmetry Topological Field Theory (SymTFT), encoding all symmetries and anomalies.
- **Physical Boundary ($\mathcal{M}_D$):** The locus of the dynamical QFT.
- **Gapped Boundary:** Implements the various presentation the symmetry structure.

SymTFT for $U(1)$ Symmetries

For a $U(1)$ 0-form symmetry in $D = 3$, the SymTFT is a 4D $\mathbb{R} \times U(1)$ BF-theory [Antinucci and Benini, 2025, Brennan and Sun, 2024]:

$$\mathcal{S}_{\text{SymTFT}} = \int_{\mathcal{Y}_4} b^{(2)} \wedge dA^{(1)} + \text{(anomaly theory)}$$

- **The $U(1)$ Field $A^{(1)}$** $\quad \& \quad W_n(\mathcal{M}_1) = e^{in \int_{\mathcal{M}_1} A}$
  When fixed on the boundary, acts as the background gauge field. Open lines endpoints define charged operators.
- **The $\mathbb{R}$ Field $b^{(2)}$** $\quad \& \quad U_\alpha(\mathcal{M}_2) = e^{i\alpha \int_{\mathcal{M}_2} b}$
  Free on the boundary; its bulk topological operators define the symmetry defects.

Dynamical Gauging of $U(1)$

Dynamical gauging converts classical background fields into dynamical DOF.

In the **SymTFT** we have a recipe [Antinucci and Benini, 2025]:

- **Bulk Replacement:** Classical background connections are replaced by unconstrained bulk field strengths:
$$dA^{(1)} \to f^{(2)} \qquad \& \qquad b^{(2)} \to dC^{(1)}$$
- **Structural Effect:** This formal shift alters the bulk-SymTFT action.

Dynamical gauging maps one symmetry structure into another, namely

$$\mathcal{S}_{\text{SymTFT}}^{\text{gauged}} = \int_{\mathcal{Y}_4} dC^{(1)} \wedge f^{(2)} + (\text{anomaly theory})$$

which in $D = 3$ happens to be again a $U(1)$ 0-form symmetry.

Fate of Symmetries and Non-Invertibility

Under dynamical gauging, due to mixed anomalies, certain global symmetries are expected to be lost. In specific occasions, they can be saved as non-invertible symmetries.

- **The Tail:** The symmetry operator develops a topological tail D_2, with $\partial D_2 = \mathcal{M}_1$:

$$\tilde{U}_\alpha(\mathcal{M}_1, D_2) = \exp\left[i\alpha\left(\int_{\mathcal{M}_1} a^{(1)} - \int_{D_2} \Omega_{\mathcal{A}}\right)\right]$$

- **Consequence:** The operator cannot be isolated on the boundary. It is then trivially contractible.

Rescuing the Non-conserved Symmetry

Resolving Non-Genuineness: The Tube Framework

To save it, we must cut-open the tail to allow non contractible $\mathcal{M}_1$.

The Idea:

- Puncture the support manifold D_2.
- Deform the support into a non-contractible higher-dimensional cylinder.
- Dress the outer boundary of this tube with a localized topological edge mode theory designed to cancel the anomaly inflow from the fields living on D_2.
- Use the topological property to shrink the cylinder into a circle.

Two Concrete Realizations

Scenario 1: The Cubic Anomaly $F_A \wedge F_B \wedge F_B$

Consider a D-dimensional QFT with the symmetries $U(1)_A^{(p-1)}$, $U(1)_B^{(r-1)}$ and SymTFT:

$$\mathcal{S}_{\text{SymTFT}} = \frac{i}{2\pi} \int_{\mathcal{Y}_{D+1}} a_A^{(D-p)} \wedge dB_A^{(p)} + a_B^{(D-r)} \wedge dB_B^{(r)} + \frac{1}{4\pi^2} B_A^{(p)} \wedge dB_B^{(r)} \wedge dB_B^{(r)}$$

This imposes the rigid dimensional constraint: $p + 2r + 1 = D$.

Gauging the $U(1)_B^{(r-1)}$ symmetry via $a_B^{(D-r)} \to dC^{(D-r-1)}$ and $dB_B^{(r)} \to f^{(r+1)}$ yields:

$$\mathcal{S}_{\text{SymTFT}}^{\text{gauged}} = \frac{i}{2\pi} \int_{\mathcal{Y}_{D+1}} a_A^{(D-p)} \wedge dB_A^{(p)} + dC^{(D-r-1)} \wedge f^{(r+1)} + \frac{1}{4\pi^2} B_A^{(p)} \wedge f^{(r+1)} \wedge f^{(r+1)}$$

Scenario 1: Bulk Equations of Motion

Varying the gauged SymTFT action yields the bulk equations of motion:

$$
\begin{aligned}
dB_A^{(p)} &= 0 \\
df^{(r+1)} &= 0
\end{aligned}
$$

$$
\begin{aligned}
-da_A^{(D-p)} + \frac{1}{4\pi^2} f^{(r+1)} \wedge f^{(r+1)} &= 0 \qquad (1) \\
dC^{(D-r-1)} + \frac{1}{2\pi^2} B_A^{(p)} \wedge f^{(r+1)} &= 0 \qquad (2)
\end{aligned}
$$

- Equation (2) establishes a generalized higher-group structure in the bulk.
- Equation (1) defines the generalized ABJ-like anomaly that triggers the non-genuineness (the tail) of the symmetry defect operator build out of $a_A^{(D-p)}$.

Scenario 1: Symmetry Defect Construction

Applying the cylinder construction at a rational angle $\alpha = \frac{2\pi}{N}$ yields the genuine localized defect operator:

$$\mathcal{D}_{\frac{1}{N}}(\mathcal{M}_{D-p}) = \int \mathcal{D}A^{(r)} \exp\left[i \int_{\mathcal{M}_{D-p}} \frac{2\pi}{2N} a_A^{(D-p)} + \frac{N}{4\pi} A^{(r)} \wedge dA^{(r)} + \frac{1}{2\pi} A^{(r)} \wedge f^{(r+1)} \right]$$

where $A^{(r)}$ is a r-form gauge field living only on the symmetry defect.

The theory we have used as topological edge mode is (higher dimensional) FQH theory.

Scenario 1: Acquiring some intuition

Varying with respect to the edge mode $A^{(r)}$ yields the localized equation of motion:

$$2\frac{N}{4\pi}dA^{(r)} + \frac{1}{2\pi}f^{(r+1)} = 0 \implies dA^{(r)} = -\frac{1}{N}f^{(r+1)} \tag{3}$$

If we locally substitute the solution of (3) ($A^{(r)} = -\frac{1}{N}b^{(r)}$ where $f^{(r+1)} = db^{(r)}$) back into the new symmetry defect, we naively recover the fractional angle defect:

$$\mathcal{D}_{\frac{1}{N}}(\mathcal{M}_{D-p}) \to \exp\left[i\int_{\mathcal{M}_{D-p}} \frac{2\pi}{2N}a_A^{(D-p)} - \frac{1}{4\pi N}b^{(r)} \wedge db^{(r)}\right]$$

This slide is wrong. The flux quantization does not match.

Scenario 1 Applied: Reduction to the 4D ABJ Anomaly

We project the general **Scenario 1** formula down to standard dimensions:

- Set $p = r = 1$, i.e. two 0-form symmetries, and $D = 4$.
- The bulk fields $a_A^{(D-p)}$ and $f^{(2)}$ reduce to a 3-form gauge field and a 2-form field strengths. The boundary edge mode $A^{(r)}$ becomes a standard 1-form connection.
- The engineered defect reduces to:

$$\mathcal{D}_{\frac{1}{N}}(\mathcal{M}_3) = \int \mathcal{D}A^{(1)} \exp\left[i\int_{\mathcal{M}_3} \frac{2\pi}{2N} a_A^{(3)} + \frac{N}{4\pi} A^{(1)} \wedge dA^{(1)} + \frac{1}{2\pi} A^{(1)} \wedge f^{(2)}\right]$$

- **Synthesis:** This recovers the exact 0-form non-invertible chiral/axial symmetry operator engineered in 4D QED [Cordova and Ohmori, 2022, Choi et al., 2022].

Scenario 2: The Mixed Anomaly $F_A \wedge F_B \wedge F_C$

We extend the idea to three distinct higher symmetries $U(1)_A^{(p-1)}$, $U(1)_B^{(r-1)}$, $U(1)_C^{(s-1)}$ with mixed anomaly, governed by the parent bulk SymTFT action:

$$\mathcal{S}_{\text{SymTFT}} = \frac{i}{2\pi} \int_{\mathcal{Y}_{D+1}} a_A^{(D-p)} \wedge dB_A^{(p)} + a_B^{(D-r)} \wedge dB_B^{(r)} + a_C^{(D-s)} \wedge dB_C^{(s)} + \frac{1}{4\pi^2} B_A^{(p)} \wedge dB_B^{(r)} \wedge dB_C^{(s)}$$

This imposes the dimensional constraint: $p + r + s + 1 = D$.

Simultaneously gauging $U(1)_B$ and $U(1)_C$ via

$$a_B^{(D-r)} \to dC_B^{(D-r-1)} \qquad a_C^{(D-s)} \to dC_C^{(D-s-1)} \qquad dB_B^{(r)} \to f_B^{(r+1)} \qquad dB_C^{(s)} \to f_C^{(s+1)}$$

yields the gauged SymTFT action:

$$\mathcal{S}_{\text{SymTFT}}^{\text{gauged}} = \frac{i}{2\pi} \int_{\mathcal{Y}_{D+1}} a_A^{(D-p)} \wedge dB_A^{(p)} + dC_B^{(D-r-1)} \wedge f_B^{(r+1)} + dC_C^{(D-s-1)} \wedge f_C^{(s+1)} + \frac{1}{4\pi^2} B_A^{(p)} \wedge f_B^{(r+1)} \wedge f_C^{(s+1)}$$

Scenario 2: Bulk Equations of Motion

Varying the gauged action leads to the bulk equations of motion:

$$dB_A^{(p)} = f_B^{(r+1)} = f_C^{(s+1)} = 0$$

$$dC_B^{(D-r-1)} + \frac{1}{4\pi^2} B_A^{(p)} \wedge f_C^{(s+1)} = 0 \tag{4}$$

$$dC_C^{(D-s-1)} + \frac{1}{4\pi^2} B_A^{(p)} \wedge f_B^{(r+1)} = 0 \tag{5}$$

$$da_A^{(D-p)} - \frac{1}{4\pi^2} f_B^{(r+1)} \wedge f_C^{(s+1)} = 0 \tag{6}$$

Equation (6) dictates the dynamical non-conservation equation, the one we are interested to save, and will turn into non-invertible.

Equations (4) and (5) are the symptoms of a higher-group structure.

Scenario 2: Multi-Field Edge Engineering

To shield the core operator, the tube protocol requires a multi-field boundary edge theory containing two interconnected higher gauge fields, $A_B^{(r)}$ and $A_C^{(s)}$:

$$\mathcal{D}_{\frac{1}{N}} = \int \mathcal{D}A_B^{(r)} \mathcal{D}A_C^{(s)} \exp\left[i \int_{\mathcal{M}_{D-p}} \frac{2\pi}{N} a_A^{(D-p)} + \frac{N}{2\pi} A_B^{(r)} \wedge dA_C^{(s)} + \frac{1}{2\pi} A_B^{(r)} \wedge f_C^{(s+1)} - \frac{1}{2\pi} A_C^{(s)} \wedge f_B^{(r+1)} \right]$$

where $A_B^{(r)}$ and $A_C^{(s)}$ are a r- and s-form gauge field living only on the symmetry defect.

The theory we have used as topological edge mode is a dimensional reduction of (higher dimensional) FQH theory.

Scenario 2: Reduction to 3D Anomalies

We project Scenario 2 onto lower dimensions to match known results:

- Set $p = 2, r = 0, s = 0$ (or $p = 1, r = 1, s = 0$) on a $D = 3$ physical boundary.
- This describes a 3D physical theory carrying a non-invertible 1-form (or 0-from) symmetry coupled to two (-1)-form gauge fields (or one (-1)-form and one 0-form).
- The engineered multi-field defect operator reduces to:

$$\mathcal{D}_{\frac{1}{N}}(\mathcal{M}_1) = \int \mathcal{D}A_B^{(0)} \mathcal{D}A_C^{(0)} \exp\left[i \int_{\mathcal{M}_1} \frac{2\pi}{N} a_A^{(1)} + \frac{N}{2\pi} A_B^{(0)} \wedge dA_C^{(0)} + \frac{1}{2\pi} A_B^{(0)} \wedge f_C^{(1)} - \frac{1}{2\pi} A_C^{(0)} \wedge f_B^{(1)} \right]$$

- **Synthesis:** This matches the exact non-invertible defect lines appeared in 3D QFT [Damia et al., 2023], and generalizes the method.

What was my motivation?

What drove me to the question: *How do I build a continuous non-invertible symmetry?*, was the desire to see if this duality web is truly complete, using generalized symmetries and gauging techniques to look for hidden gaps.

Conclusions

Conclusions

What happened?

- Introduced Generalized Symmetries
- Convinced you they are interesting
- Introduced the SymTFT to talk about the symmetry structure of a QFT
- Developed two cases in which we understand the formation of continuous non-invertible symmetry after gauging

Thank you for your attention!

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Backup Slides

Scenario 2: Get some intuition again

Varying this engineered defect action with respect to $A_B^{(r)}$ and $A_C^{(s)}$ yields the constraints:

$$\frac{N}{2\pi} dA_C^{(s)} + \frac{1}{2\pi} f_C^{(s+1)} = 0 \qquad \text{and} \qquad -\frac{N}{2\pi} dA_B^{(r)} + \frac{1}{2\pi} f_B^{(r+1)} = 0$$

substitute ($A_C^{(s)} = -\frac{1}{N} b_C^{(s)}$ and $A_B^{(r)} = \frac{1}{N} b_B^{(r)}$) in the well defined symmetry defect

$$\mathcal{D}_{\frac{1}{N}}(\mathcal{M}_{D-p}) \to \exp\left[i \int_{\mathcal{M}_{D-p}} \frac{2\pi}{N} a_A^{(D-p)} - \frac{1}{2\pi N} b_B^{(r)} \wedge db_C^{(s)} \right]$$

Remember: this slide is wrong again.

Scenario 1 & 2: Extension to -1-Form Symmetries

In both cases we can extend to -1-form symmetries (and -2-form), some examples:

- (S1) $p = r = 0 \implies D = 1$: Generates a non-invertible -1-form symmetry in 1D, i.e. Quantum Mechanics.
- (S1) $p = 3, r = 0 \implies D = 4$: 2-form non-invertible symmetry realized in 4D.
- (S2) $p = 1, r = -1, s = 1 \implies D = 2$: Realized via generalized 0-form symmetry structures [Lin et al., 2025]

$U(1)$ Dynamical Gauging: Bulk Action

Photon Sector: The dynamical 1-form bulk sector is governed by:

$$S_{\text{photon}} = \frac{i}{2\pi} \int_{\mathcal{Y}_{d+1}} \left(g^{(d-2)} \wedge dG^{(2)} + f^{(2)} \wedge dF^{(d-2)} - G^{(2)} \wedge dF^{(d-2)} \right)$$

- **Bulk Stitching:** To realize minimal coupling, bulk surfaces of $G^{(2)}$ must terminate directly on the lines of $A^{(1)}$.
- **Total Action:** Synthesized via the non-derivative topological linking term $b^{(d-1)} \wedge G^{(2)}$:

$$S_{\text{total}} = \frac{i}{2\pi} \int_{\mathcal{Y}_{d+1}} \left(b^{(d-1)} \wedge dA^{(1)} + b^{(d-1)} \wedge G^{(2)} + S_{\text{photon}} \right)$$

$U(1)$ Dynamical Gauging: Non-Genuineness

- **Modified Gauge Algebra:** The topological linking forces non-trivial gauge transformations:

$$\delta G^{(2)} = d\lambda^{(1)} \qquad \text{and} \qquad \delta A^{(1)} = -\lambda^{(1)}$$

- **Non-Genuine Operators:** Isolated lines of $A^{(1)}$ lose gauge invariance, requiring attachment to a bounding bulk disk $\mathcal{D}_2$:

$$L_n(\gamma_1, \mathcal{D}_2) = \exp\left(in \int_{\gamma_1} A^{(1)} + in \int_{\mathcal{D}_2} G^{(2)} \right), \qquad \partial\mathcal{D}_2 = \gamma_1$$

- **Dual Screening:** Variations of $b^{(d-1)}$ trivialize and break the continuous $U(1)^{(0)}$ defects.
- **Effective IR Map:** Integrating out trivialized sectors recovers the dual QED SymTFT via the formal substitution:

$$dA^{(1)} \longrightarrow f^{(2)} \qquad \text{and} \qquad b^{(d-1)} \longrightarrow dF^{(d-2)}$$